

ALFIDO TECH INTERNSHIP

Task 2: Console-Based Mini Project

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Internship: Java Developer Internship

Task Number: 2

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Objective :

The objective of this task is to develop a console-based Java application using Scanner for user input. The program demonstrates loops, conditional statements, and basic Java programming concepts through a Student Management System.

Programs Implemented :

Student Management System

Description :

This application allows users to manage student information using a console interface. It accepts user input using the Scanner class and demonstrates loops and conditional logic.

Features :

- Uses Scanner for user input
- Uses loops
- Uses if-else conditions
- Console-based application
- Displays student details

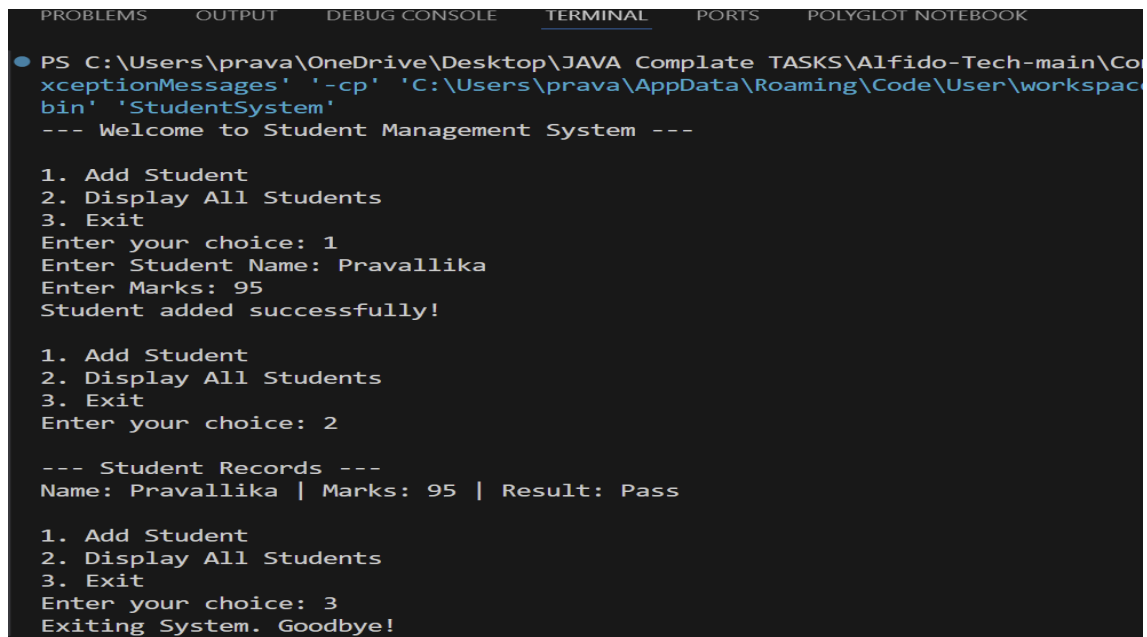
Source Code :

```
J StudentSystem.java X
J StudentSystem.java
1  import java.util.Scanner;
2
3  public class StudentSystem {
4      Run | Debug
5      public static void main(String[] args) {
6          Scanner sc = new Scanner(System.in);
7
8          // Arrays to store student data
9          String[] names = new String[5];
10         int[] marks = new int[5];
11         int studentCount = 0;
12
13         System.out.println(x: "--- Welcome to Student Management System ---");
14
15         while (true) {
16             System.out.println(x: "\n1. Add Student");
17             System.out.println(x: "2. Display All Students");
18             System.out.println(x: "3. Exit");
19             System.out.print(s: "Enter your choice: ");
20
21             int choice = sc.nextInt();
22             sc.nextLine(); // Consume newline
23
24             if (choice == 1){
25                 if (studentCount < 5) {
26                     System.out.print(s: "Enter Student Name: ");
27                     names[studentCount] = sc.nextLine();
28
29                     System.out.print(s: "Enter Marks: ");
30                     marks[studentCount] = sc.nextInt();
31                     sc.nextLine(); // IMPORTANT FIX
32
33                     studentCount++;
34                     System.out.println(x: "Student added successfully!");
35                 } else {
36                     System.out.println(x: "System Full! Cannot add more students.");
37                 }
38             }
39         }
40     }
41 }
```

```
J StudentSystem.java X
J StudentSystem.java
3  public class StudentSystem {
4      public static void main(String[] args) {
5
39         else if (choice == 2) {
40             if (studentCount == 0) {
41                 System.out.println(x: "No records found.");
42             } else {
43                 System.out.println(x: "\n--- Student Records ---");
44                 for (int i = 0; i < studentCount; i++) {
45                     String result = (marks[i] >= 35) ? "Pass" : "Fail";
46                     System.out.println("Name: " + names[i] +
47                                         " | Marks: " + marks[i] +
48                                         " | Result: " + result);
49                 }
50             }
51         }
52
53         else if (choice == 3) {
54             System.out.println(x: "Exiting System. Goodbye!");
55             break;
56         }
57
58         else {
59             System.out.println(x: "Invalid choice! Please try again.");
60         }
61     }
62
63     sc.close();
64 }
65 }
```

Figure 1: StudentSystem.java Source Code

Output :



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POLYGLOT NOTEBOOK
● PS C:\Users\prava\OneDrive\Desktop\JAVA Complate TASKS\Alfido-Tech-main\Co
xceptionMessages' '-cp' 'C:\Users\prava\AppData\Roaming\Code\User\workspac
bin' 'StudentSystem'
--- Welcome to Student Management System ---

1. Add Student
2. Display All Students
3. Exit
Enter your choice: 1
Enter Student Name: Pravallika
Enter Marks: 95
Student added successfully!

1. Add Student
2. Display All Students
3. Exit
Enter your choice: 2

--- Student Records ---
Name: Pravallika | Marks: 95 | Result: Pass

1. Add Student
2. Display All Students
3. Exit
Enter your choice: 3
Exiting System. Goodbye!
```

Figure 2: StudentSystem Output

Conclusion :

This task helped me understand how to build console-based Java applications using Scanner, loops, and conditional statements.